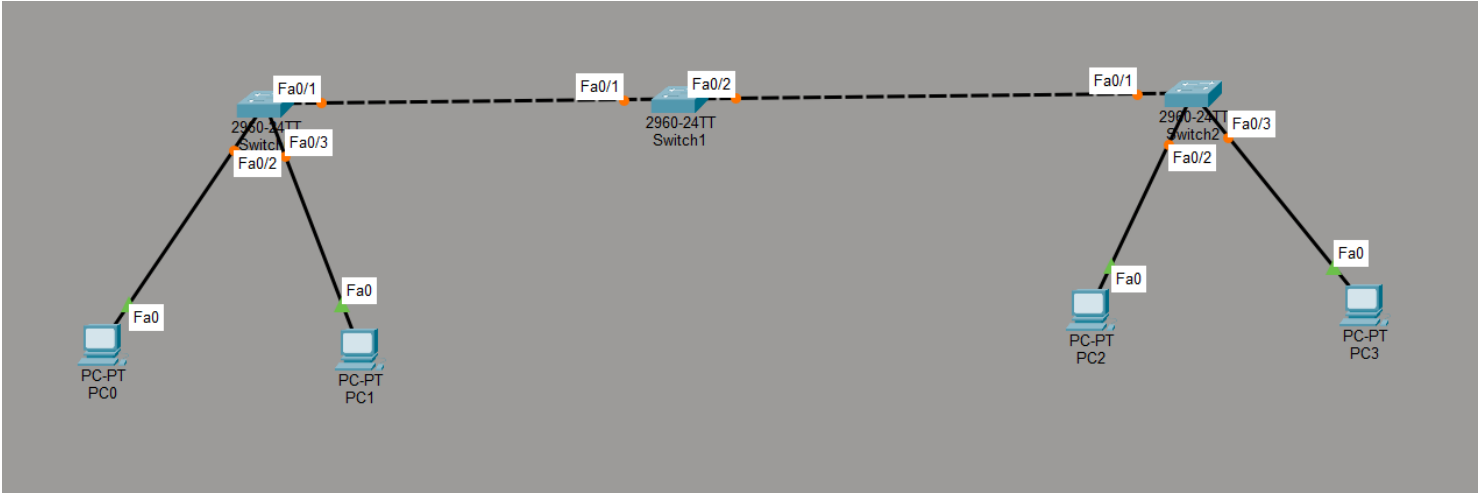


Q- On Cisco Packet Tracer, create a network with three switches and assign at least two different VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support'). Connect two PCs to each switch and assign each PC to the correct VLAN based on department.

To create a network using three switches, configure two VLANs (VLAN 10 for Sales and VLAN 20 for Support), connect two PCs to each switch where applicable, and verify communication between devices belonging to the same VLAN.



IP Addressing

PC	Department	VLAN	IP Address	Subnet Mask
PC0	Sales	VLAN 10	192.168.10.1	255.255.255.0
PC1	Support	VLAN 20	192.168.20.1	255.255.255.0
PC2	Sales	VLAN 10	192.168.10.2	255.255.255.0
PC3	Support	VLAN 20	192.168.20.2	255.255.255.0

Switch 0

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Support
Switch(config-vlan)#exit
Switch(config)#int f 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#int f 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int f 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#exit
```

Switch 1

```
Switch(config)#int f 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#int f 0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch(config-if)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
```

Switch 2

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vl
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
% Ambiguous command: "v"
Switch(config)#vlan 10
Switch(config-vlan)#name Sales
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Support
Switch(config-vlan)#exit
Switch(config)#int f 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#int f 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int f 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
```

Ping Testing

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```